Biotechnology

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BIOTECHNOLOGY

1. TOOLS
Restriction enzymes (molecular scissors)
DNA ligase
Vectors: plasmids, viruses
Host: E. coli, yeast

2. TECHNIQUES
PCR: amplify DNA
Denature 95 C
Anneal 55 C
Extend 72 C
Gel electrophoresis: separate DNA
DNA sequencing: Sanger method
CRISPR-Cas9: gene editing

3. APPLICATIONS
Medicine:
Insulin production
Vaccines
Gene therapy
Monoclonal antibodies
Agriculture:
GM crops (Bt cotton, Golden rice)
Pest resistance
Environment:
Bioremediation
Biofuels

4. GENOMICS
Human Genome Project
Bioinformatics
Personalized medicine

5. ETHICS
GMO concerns
Gene editing (germline)
Cloning
Privacy of genetic data